

Lecture: Clinical Decision Making

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Do.
Make.
Innovate.
Reinvent the Future.

Lecture OBJECTIVES

At the conclusion of the session students should be able to:

1. Define Dual Process Theory of Cognition (System 1 and System 2) as the leading model for explaining clinical decision making
2. Apply concepts of problem representation, illness scripts and pattern recognition
3. Demonstrate knowledge and application of diagnostic reasoning process.
4. Distinguish between the different types of Clinical cognitive errors based on a clinical vignette.

- Source: Course Syllabus

As physicians after we **gather** a history and **perform** an appropriate physical exam based on the reason for the patient seeking care we must:

- **Analyze** the findings to identify potential causes for the patient's symptoms using **clinical reasoning**
- The potential causes become our Differential Diagnosis (DDX)
 - List starting with most likely and including other plausible options that explain the patient symptoms
 - List may include diagnoses that have serious consequences if missed.
 - Example: Patient with acute chest pain radiating to the left shoulder should be thought of as an myocardial infarction until proven otherwise because of the potential lethality if not diagnosed and treated immediately

Clinical Decision Making

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“Thinking about how you make clinical decisions is the first step toward making better ones”

- Donald R. Woolever, MD. The Art and Science of Clinical Decision Making. Family Practice Management. May 2008.

HEURISTICS

- **Mental shortcuts**
 - Used to assist clinician to rapidly make decisions (System 1 analytics)
 - Generally subconscious formulated response patterns based on formed habits.
 - Subject to error because of all our cognitive biases.
-
- Croskerry P. From mindless to mindful practice –cognitive bias and clinical decision making. N Engl J Med. 2013;368(26):2445-2448.

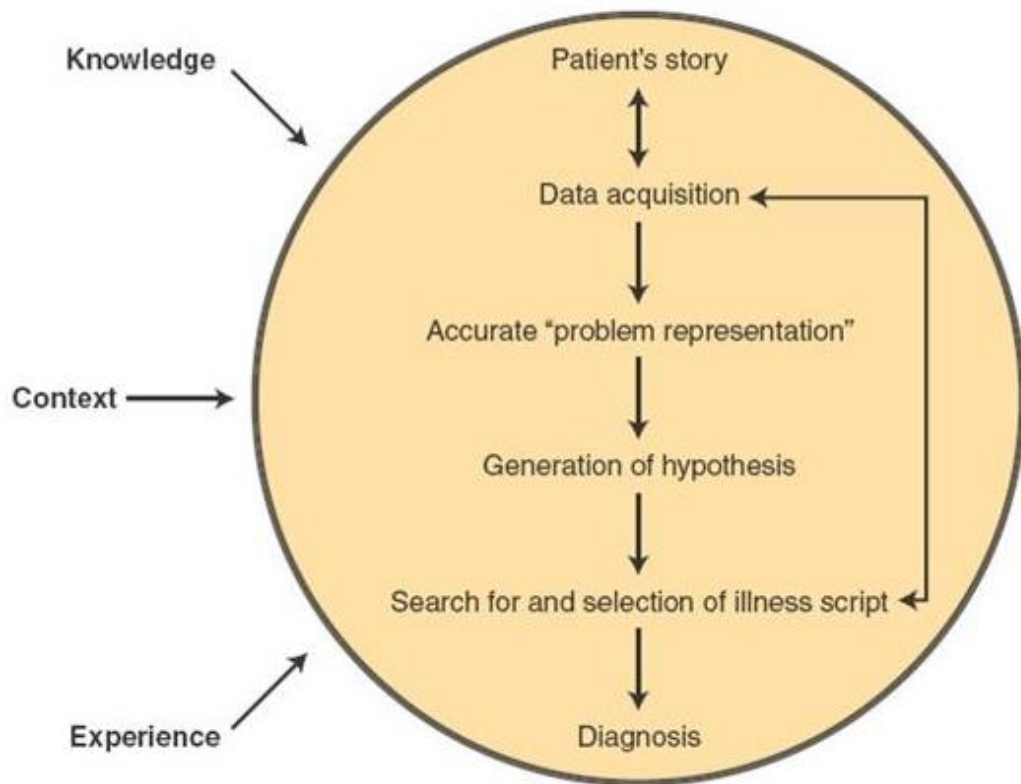
Clinical Decision Making: Example

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These are all pictures of what object?

FIGURE 5-1



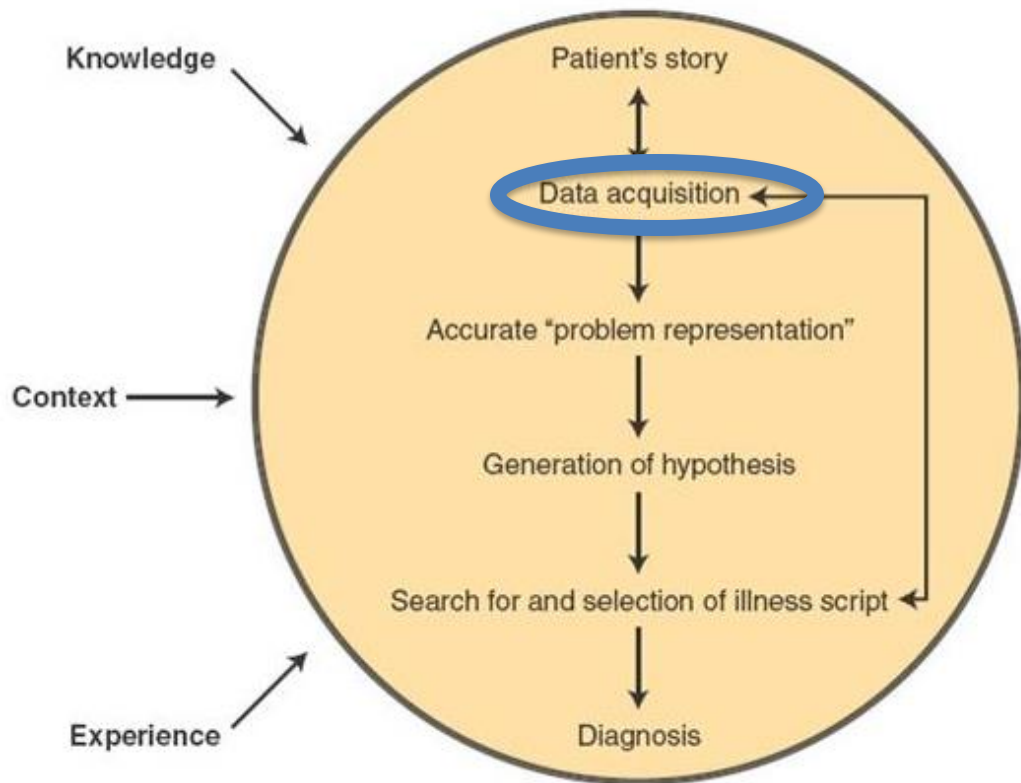
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Key elements of the clinical diagnostic reasoning process.

(From Bowen JL. *N Engl J Med*. 2006;355(21):2217–2225. Copyright © 2006 Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.)

- Gathering Initial Health history and Physical Exam
 - Symptom's elicited during the history
 - Signs- normal or abnormal observed during the physical exam
 - Laboratory or diagnostic reports
 - Be organized & methodical so all abnormal and unexpected findings are identified

FIGURE 5-1



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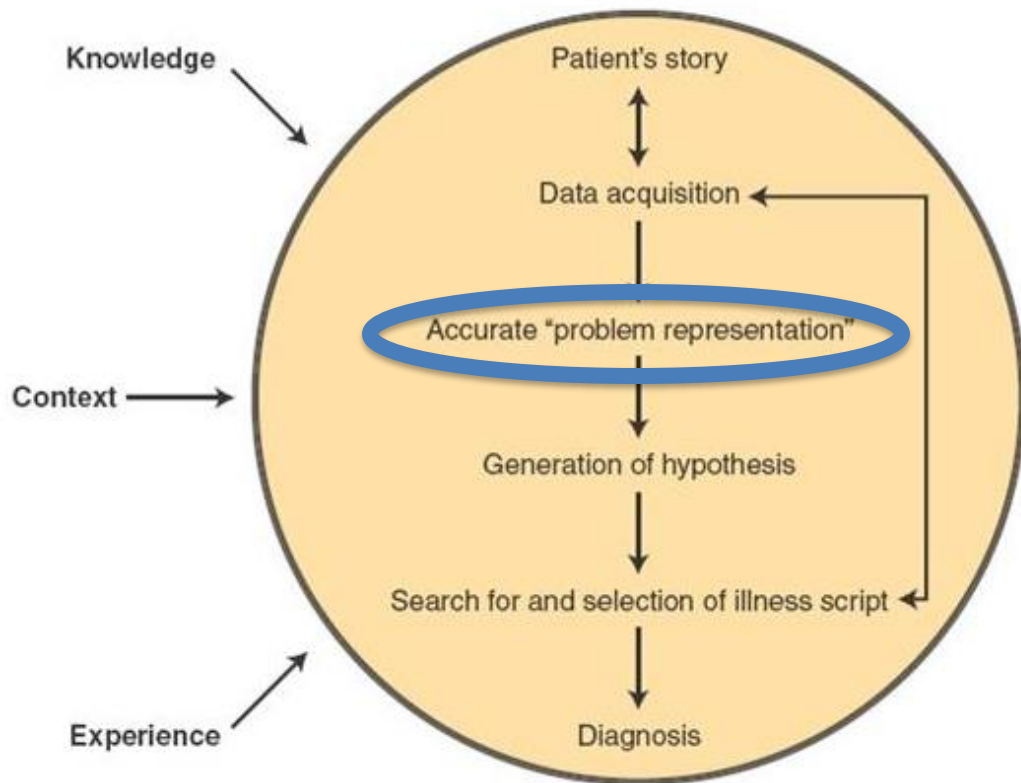
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Approaches to organizing findings:

1. Anatomic Location- cluster findings based on their anatomic relationship in the body.
2. Age-What disease processes are common in different age groups;
3. Timing of Symptoms-knowing how different disease progress as well as duration of certain diseases is important.
4. Involvement of body systems
5. Multisystem conditions

FIGURE 5-1



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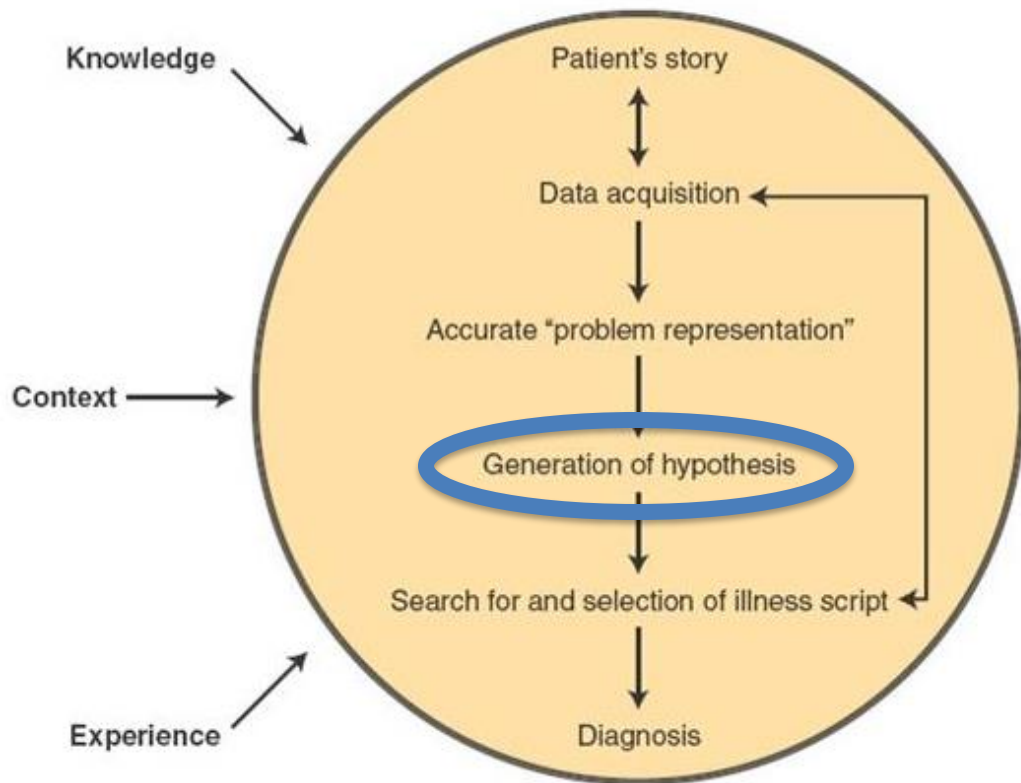
Synthesizing Information and Developing the Problem Representation

- You make sense of all the information gathered to this point and form a problem representation or clinical picture about the patient.
- Includes:
 - Initial information (CC, epidemiology, risk factors)
 - Key features in the H & P
 - Results of diagnostic testing

*In your clinical documentation, the problem representation is called the summary statement that often starts the assessment section of your note.

Please review box 5-2 in Bates 13ed and try writing a problem representation of your own.

FIGURE 5-1



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Generating Hypotheses (Differential Diagnosis)

- For each identified problem or cluster of problems, you will generate a clinical hypothesis for the underlying cause or diagnosis.

BOX 5-3

Approaches to Searching for Probable Causes of the Findings

- Generate an exhaustive list
- Match findings against all conditions that can produce them
- Eliminate diagnostic possibilities that fail to explain the findings
- Weigh competing possibilities and select the most likely diagnosis
- Give special attention to potentially life-threatening conditions

Memory Aid For Generating Differential Diagnosis (exhaustive method)

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TOM G PRINCE, MD, Psychiatrist, General Hospital

Toxins/Trauma including meds

Oncologic

Msk/Rheumatologic

Gastrointestinal

Pulmonary

Renal

Infectious

Neurologic

Cardiovascular

Endocrine

Metabolic

Dermatologic

Psychiatric

Genitourinary/Gyn

Hematologic

VINDICATE

Vascular

Infectious

Neoplastic

Drug related

Inflammatory/Idiopathic/Iatrogenic

Congenital

Autoimmune/Allergic

Trauma/Toxic

Endocrine/Metabolic

Generating Hypotheses (Differential Diagnosis)

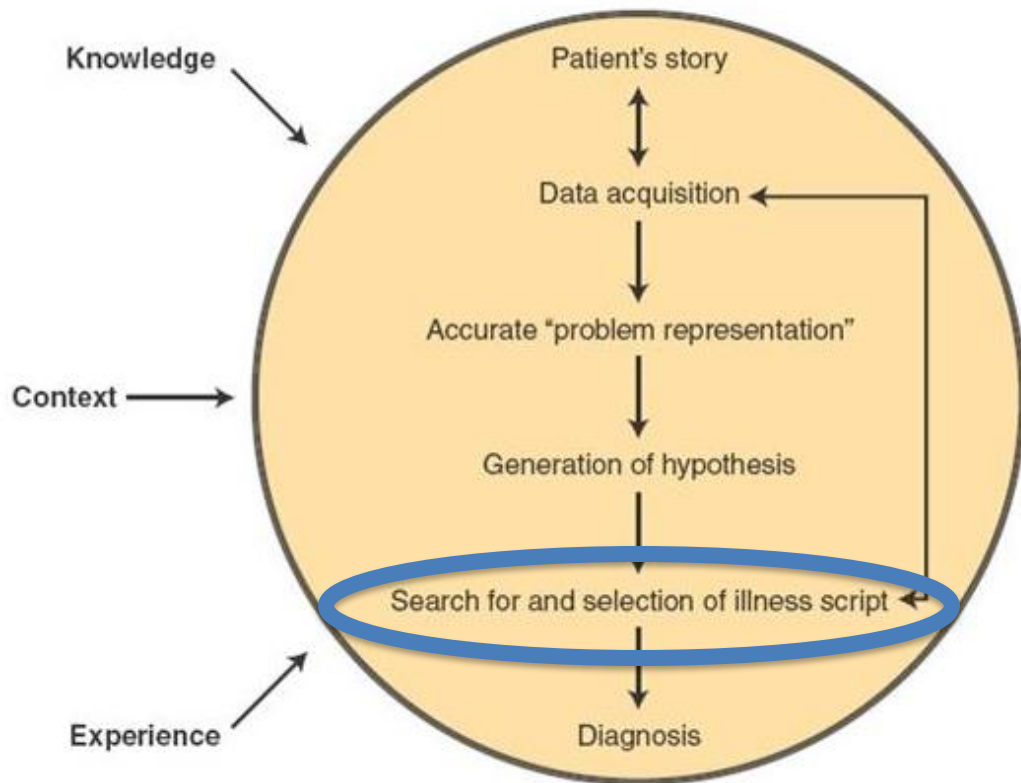
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Illness Scripts

- Organized mental summary of a clinician's knowledge of a disease
- Illness scripts often include a disease's
 - Pathophysiology
 - Epidemiology
 - Time course
 - Salient symptoms and signs
 - Diagnostics
 - Treatment

Illness Scripts. JGIM. Clinical reasoning exercises

Let's go through an example of this...

Close your eyes and picture a fire truck?
What does it look like? What color is it?

Is this what you pictured?

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What if I showed you this picture instead and asked you to tell me what this was?

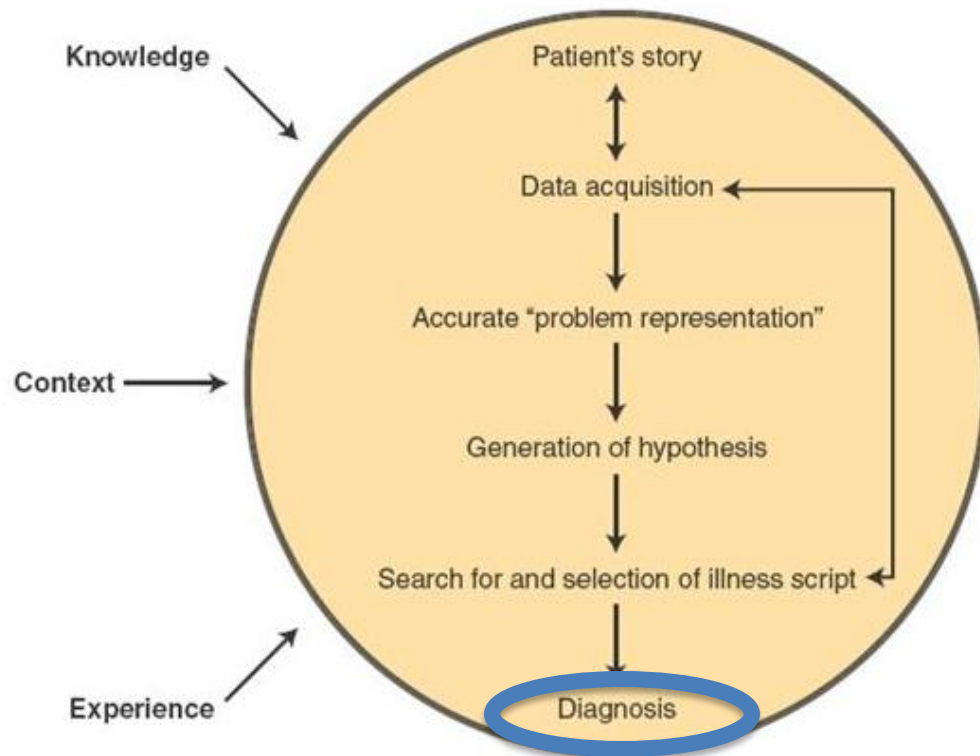
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Components of Illness Script	Community Acquired Pneumonia
Epidemiology/ Pathophysiology	Increased risk with: Age Post URI Structural Lung disease Immunodeficiency
Time course	Acute: days Progressively worsens if not treated
Salient Symptoms & Signs	Fever, cough, SOB, Tachycardia, Tachypnea, Hypoxemia
Diagnostics	Labs and imaging Leukocytosis Lobar infiltrate on CXR bacteria in sputum or blood cultures
Treatment	Antibiotics typically lead to improvement over days

- **Illness Scripts.** JGIM. Clinical reasoning exercises

FIGURE 5-1



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- From a documentation standpoint, you will list your diagnoses in the assessment section of the note.
- The diagnosis may be very straight forward or more challenging and require further information to get to a final diagnosis if there are 2 possible causes for the patient's condition.
- You always want to be specific as the data will allow.
 - Example: subarachnoid hemorrhage, unspecified vs subarachnoid hemorrhage, left temporal lobe

Planning the Diagnostic and Treatment Strategy

- The Plan of Care should be a logical flow from your working diagnoses listed in the assessment section of your note.
- The Plan should include:
 - **Laboratory tests** to allow you differentiate between 2 potential diagnoses or guide your therapeutic management of a patient.
 - **Diagnostic interventions** to help you rule in or rule out a diagnosis.
 - **Patient education**
 - **Medication changes**
 - **Referrals**
 - **Follow up visits**

Diagnostic Errors

- Diagnostic errors have been implicated in 15% of clinical decisions with medicolegal repercussions
- Misdiagnosis has the potential to cause serious harm to patients as it sets up a cascade of subsequent mistreatment
- General medical Council and other entities advocate that Med Students need to be equipped with critical thinking skills in order to identify their own cognitive biases and those of the clinicians they rotate with.

Bhatti A. Cognitive bias in clinical practice-nurturing health skepticism among medical students. *Advances in Medical Education and Practice* 2018; 9(1):235-237

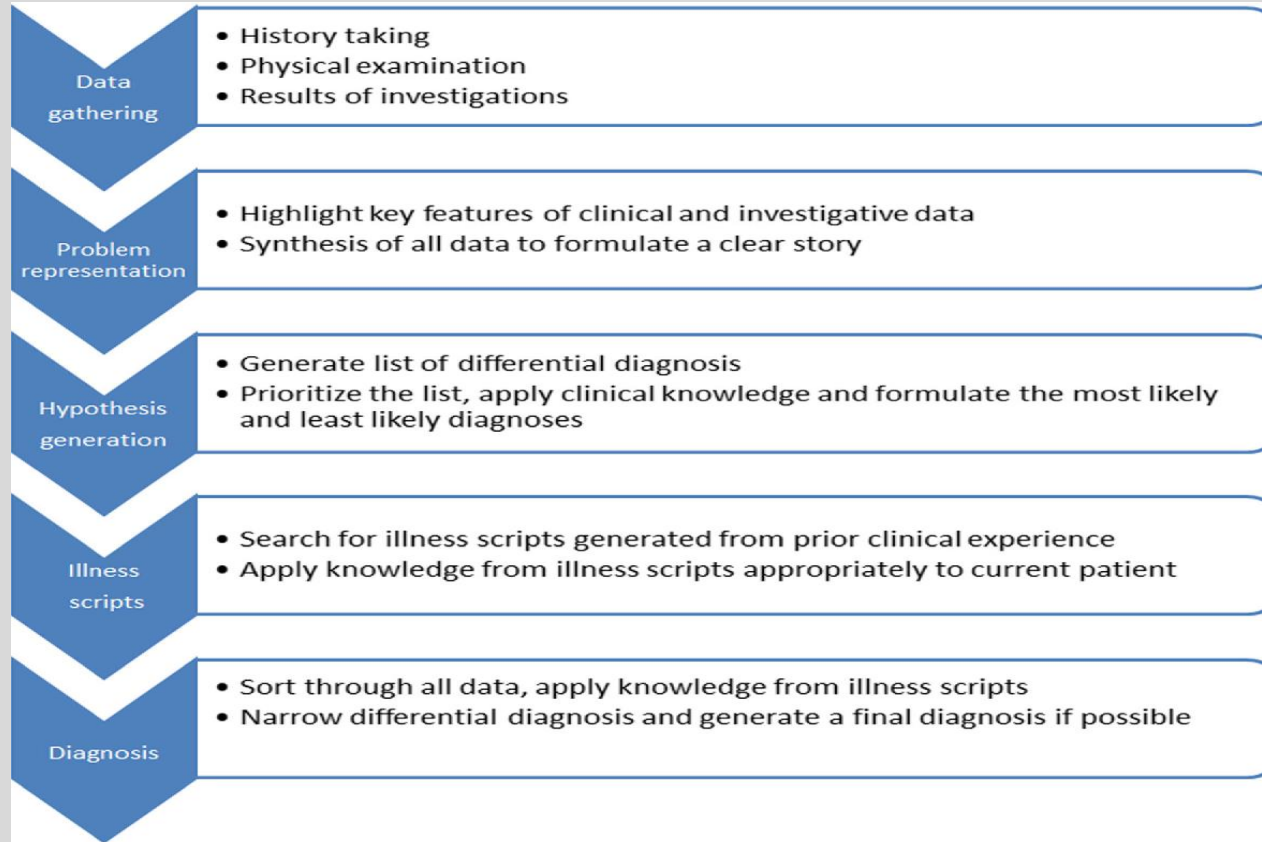
Common Diagnostic Errors

Common Types of Clinical Cognitive Errors

Cognitive Error	Description	Vignette
Anchoring bias	Tendency to perceptually lock onto salient features in the patient's initial presentation too early in the diagnostic process and failure to adjust in light of later information	A clinician "locks onto" a patient's description of an aura that precedes her headaches as indicative of a migraine and fails to recognize red flags of increased intracranial pressure that should prompt neuroimaging for this patient
Availability heuristic	Assumption that a diagnosis is more likely, or more frequently occurring, if it more readily comes to mind	A clinician who has recently seen several patients with acute appendicitis does not consider ovarian torsion in an adolescent girl presenting with acute right lower quadrant abdominal pain
Confirmation bias	Seeking supportive evidence for a diagnosis at the exclusion of more persuasive information refuting it	A clinician makes a presumptive diagnosis of an upper respiratory infection in a well-appearing patient presenting with cough, rhinorrhea, and fever, and does not consider pneumonia even after finding asymmetric chest wall excursion and dullness to chest percussion on examination
Diagnostic momentum	Prioritizing a diagnosis made by prior clinicians, discounting evidence of alternative explanations	A clinician does not consider acute myocardial infarction in a patient who was recently diagnosed with acid reflux in the setting of similar symptoms
Framing effect	Interpretation of information is influenced heavily by the way in which information about the problem are presented (<i>framed</i>)	A patient is presented as having "frequent emergency room visits for asthma exacerbation in the setting of medication noncompliance." The clinician fails to explore structural forces that drive medication adherence and fails to explore alternative causes of the current exacerbation
Representation error	Failure to take prevalence into account when estimating the probability of a diagnosis	Clinician who often sees older patients places diverticular bleed high on her differential diagnosis when evaluating rectal bleeding in an adolescent patient
Visceral bias	Visceral arousal (negative and positive feelings toward patients) lead to poor diagnostic decisions	Clinician assumes that a patient who is homeless will not be able to manage a complicated treatment plan and prescribes a simpler, less optimal plan, without discussing the options with the patient

Sources: Croskerry P. *Acad Med*. 2003;78(8):775-780; Weinstein A et al. *MedEdPORTAL*. 2017;13:10650.

Review: Steps and Strategies for Clinical Reasoning



- Bowen JL. Educational strategies to promote clinical diagnostic reasoning. *N Engl J Med*.2006;355(21):2217-25.

Let's Practice

You may want a blank sheet of paper to write
your answers on

Clinical Case

A 3 yo Asian male presents with 1 day history of fever of 102.5 F, increased irritability, and pulling at his ears. Dad states that he has noticed runny nose that started 2-3 days ago. Denies any changes in eating or drinking, changes in number of stools or wet diapers/day. States there have been kids at his daycare were out sick at the beginning of the week.

PMH: previously health; delivered by SVD at 38 weeks gestation;

Vaccination status: up to date

PSH: none

FMH: older sister h/o PET tubes at age 4 d/t recurrent acute otitis media

SH: lives at home with mom, dad, and older sister; goes to Happy hills daycare during week, no pets in home, not exposed to smokers in the home

Meds: none

Allergies: none

What conditions can cause a patient to have ear pain?

Hypothesis of potential causes (Differential Diagnosis)

Serous Otitis media

Suppurative Otitis media

Otitis externa

Eustachian tube dysfunction

Viral upper respiratory tract infection

Lymphadenitis

Temporomandibular joint disease

Internal rotation of the temporal bone

Physical Exam:

Constitutional: age appropriate male who appears ill and is clinging to dad in the office;

Eyes: conjunctiva is pink and sclera non-icteric. EOMI b/l

Ears: ear canals patent and flesh colored b/l with bulging TM and loss of middle ear architecture

Nose: nasal mucosa is erythematous with naso-turbinate edema and thin clear nasal discharge present b/l

Throat: moist buccal mucosa in oral cavity; posterior pharynx erythematous with +2 tonsils bilaterally; normal motion of uvula

Lymphatic: Shotty anterior lymphadenopathy

Cardiac: RRR; no murmurs/gallops/rubs

Resp: CTAB; no wheeze/rhales/rhonchi

MSK: cervical: increased cervical hypertonicity b/l; OA: FRrSI; thoracic: thoracic inlet: FRISr; increased paraspinal hypertonicity

Integument: No rashes, lesion noted on anterior/posterior chest

How would you write a problem representation for this patient?

Problem Representation Example:

A 3 yo Asian male who has no underlying medical conditions presents with a 2-3 day history of rhinorrhea that has progressed in the last 24 hour with 102.5 F fever and ear pain without associated diet or stooling changes. His exam is notable for normal hydration status as evidence by moist buccal mucosa and associated erythematous/edematous posterior pharynx, bulging TM b/l, anterior cervical lymphadenopathy and nasal drainage.

Lets go back to your hypotheses (Differential Diagnosis)- what might you need to include or exclude from this list? Why?

Serous Otitis media

Suppurative Otitis media

Otitis externa

Eustachian tube dysfunction

Viral upper respiratory tract infection

Lymphadenitis

Temporomandibular joint disease

Internal rotation of the temporal bone

Lets go back to your hypotheses (Differential Diagnosis)- what might you need to include or exclude from this list? Why?

~~Serous Otitis media~~ - This child has a bulging TM and you can't see the middle ear bones
Suppurative Otitis media
~~Otitis externa~~ -The ear canal was flesh colored and there was not a mention of edema in the canal
Eustachian tube dysfunction
Viral upper respiratory tract infection
Lymphadenitis
~~Temporomandibular joint disease~~ -This is an unlikely diagnosis for this age group-more common in teens and older in life as well as more common in women.
Somatic dysfunction of the head region
Streptococcal Pharyngitis/Bacterial Pharyngitis -May be a consideration given patient's pharyngitis
Conductive hearing loss

There be more than 1 diagnosis to explain this patient's symptoms?

Build an Illness Script for Suppurative Otitis Media

[Article on Otitis Media for your reference:](https://www.aafp.org/afp/2013/1001/p435.html)

[https://www.aafp.org/afp/2013/1001/p435.ht
ml](https://www.aafp.org/afp/2013/1001/p435.html)

Illness Script Example:

Suppurative Otitis Media

Epidemiology/Pathophysiology

Younger age
Risk factors include craniofacial abnormalities, allergies, exposure to smoke or respiratory irritant, daycare exposure, FMH recurrent otitis media, GERD, immunodeficiency, bottle feeding, pacifier use, URI. Common bacteria isolates include streptococcus pneumoniae, Haemophilus influenza, and Moraxella catarrhalis

Time Course

48 hours

Clinical Presentation

Ear pain, irritability, or fever

Diagnostics

Moderate to severe TM bulging, new onset otorrhea not caused by otitis externa, mild TM bulging associated with ear pain or erythema

DOES THIS FIT THE SIGNS AND SYMPTOMS OF YOUR PATIENT IN THE CASE?

What is your final diagnosis or diagnoses?

Working Diagnosis Example

Novice	Experienced Physician
1. Acute bilateral otitis media	1. Acute bilateral suppurative otitis media
2. URI	2. URI; cause unknown
	3. Bilateral Eustachian tube dysfunction
	4. Somatic dysfunction of the head region
	5. Somatic dysfunction of the thoracic region

*Note that even among experienced physicians, this Diagnosis list may be different depending specialty.

What would your plan be for this patient?

1. Perform tympanogram to assess for potential hearing involvement
 2. Start on Amoxicillin 90 mg/kg/d x 10 days
 3. Discussed repopulating GI tract with healthy bacteria and recommended feeding child yogurt once daily while on antibiotics at least 2 hours after medication to help with this.
 4. Discussed importance of keeping child hydrated and discussed warning signs including decrease in # wet diapers to follow up immediately or call the office
 5. OMT was performed to the above regions using MFR and cranial. Patient tolerated procedure well.
 6. Patient to f/u if symptoms do not start to improve in 48-72 hours.
- *If a patient has persistent AOM or if you are monitoring, you may f/u to check for resolution of middle ear fluid and or recheck ear exam.

Lecture and Lab Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>